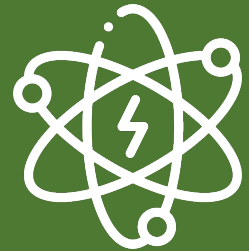
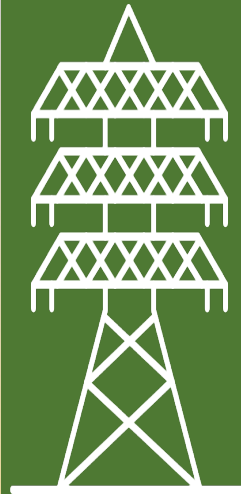
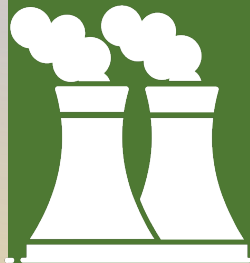


ANALYTICAL REVIEW AND YOUTH & STAKEHOLDER ENGAGEMENT ON GHANA'S ENERGY TRANSITION PLAN

CENTRE FOR EXTRACTIVES & DEVELOPMENT,
AFRICA (CEDA)

SHIKA AKPALOO

MAY 2024





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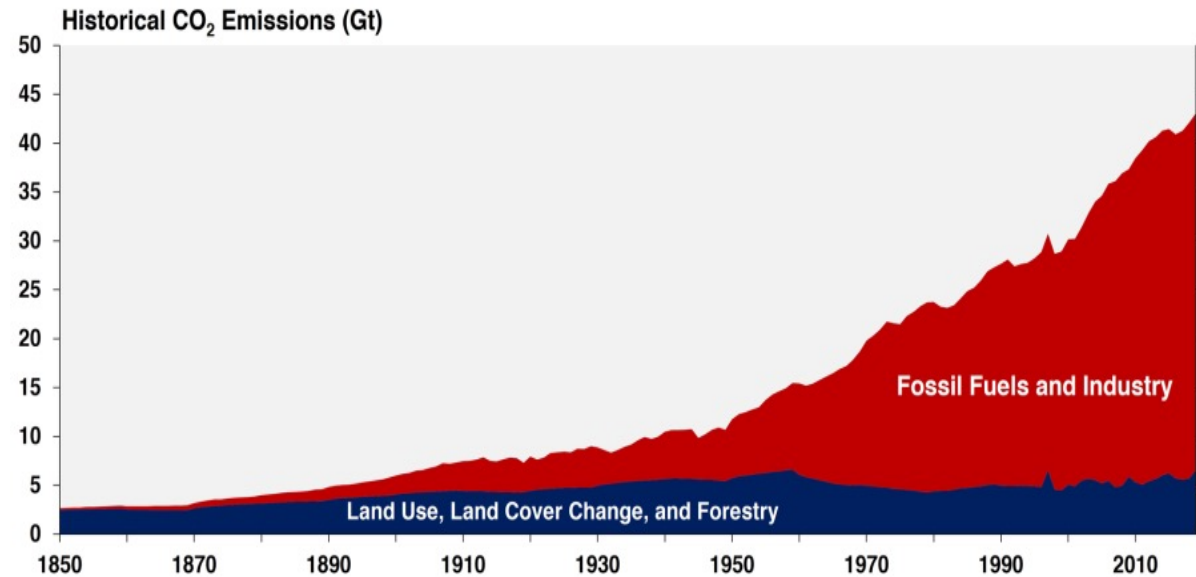
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Recommendations

Background and Context



Coal and oil and gas (fossil fuels) have powered industrialization and accelerated rapid socio-economic growth and development of countries since the 17th century.



Combustion of these fossil fuels to generate the vital energy needed to power global economies has increased greenhouse gases (GHGs) to about 92% of global CO₂ emissions in 2019 alone.

Global Energy Transition

UNITED STATES	• Approval of the inflation reduction act ○ Tax credit extensions for solar & wind ○ Investment tax credit available for battery storage & zero-emission nuclear ○ Financial support for grids & manufacturing clean power equipment	INDIA	• Continues to expand the production-linked incentive (PLI) scheme ○ 50 GWH of battery manufacturing capacity ○ 40 GW of solar PV manufacturing capacity to be added in next three years
CHINA	• 14th five-year plan raises renewable target to 33% of power consumption by 2025 (and 18% for non-hydro renewables)	JAPAN	• Government is studying extension to lifetime of nuclear power plants (beyond 60 years)
EUROPE	• Net-zero industry act proposal and other potential reforms ○ Increase EU 2030 renewables target to 45% by 2030 ○ Proposed reform of market design and technology-specific targets for EU manufacturing capacity • Nine European countries committed to boost offshore wind capacity to over 120 GW by 2030 and over 300 GW by 2050	KOREA	• Plan to increase nuclear power to 35% of total generation and renewables to 31% from 10% in 2021 by 2036 • Coal-fired power to reduce to 15%
		SOUTH AFRICA	• Government concluded sixth renewable auction
		BRAZIL	• Planning two major transmission auctions in 2023, including the largest ever held in Brazil (in investment terms)

Objectives of this Study



Assess the adequacy & feasibility of Ghana's energy transition targets, policies, & measures



Evaluate the socio-economic implications of the energy transition



To examine the financial & investment landscape for the energy transition



To analyze the institutional readiness and capacity of key sectors,



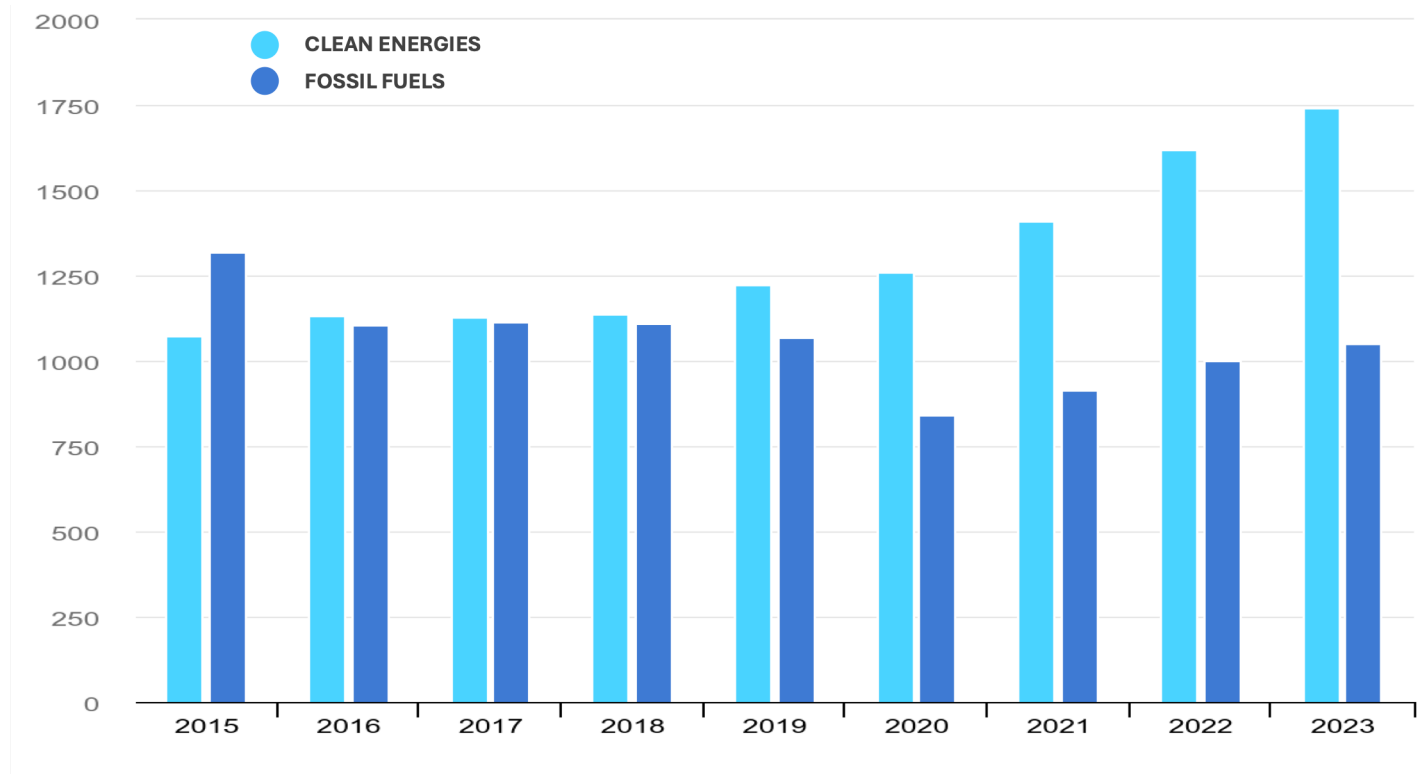
To explore the role of critical minerals & associated challenges and opportunities in their sustainable exploitation.



Establish the relevance of youth engagement & identify opportunities for youth in the transition process

Global Energy Investment Landscape

Global investment in renewable energy estimated to have increased from **1074 billion USD in 2015** to **1740 billion USD in 2023**



Investment in fossil fuels decreased from **1319 billion USD** to **1050 billion USD in 2023**

Source: IEA (2023) Global Energy Investment in Clean Energy and in Fossil Fuels, 2015-2023

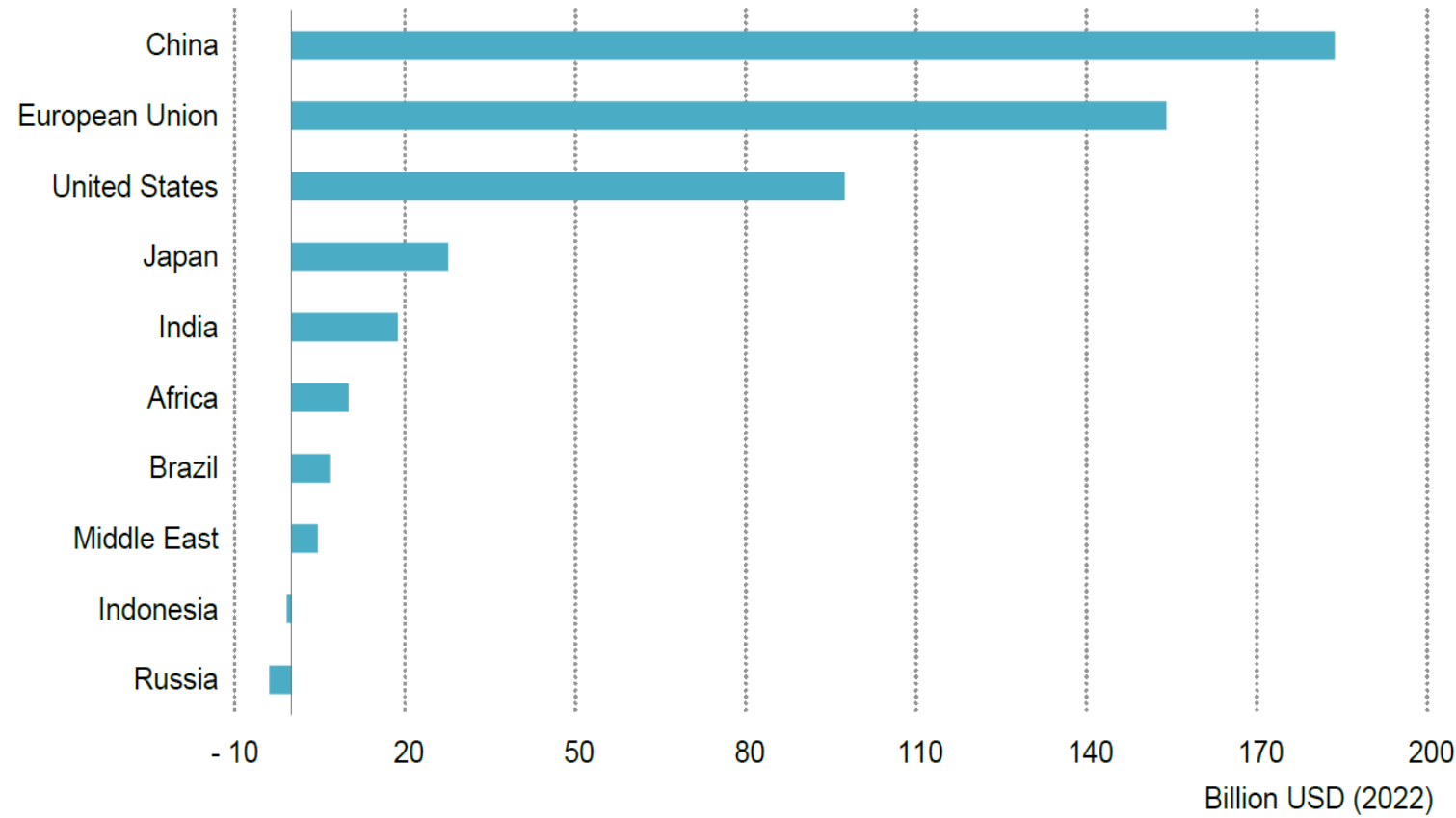
Global Energy Investment Landscape

Increase in clean energy spending is concentrated in developed countries

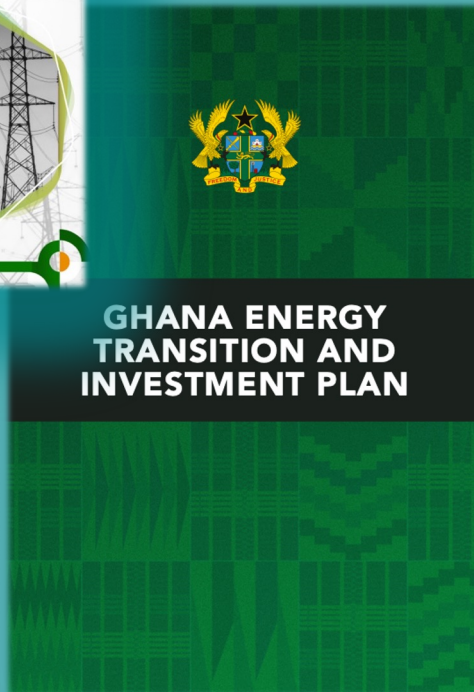
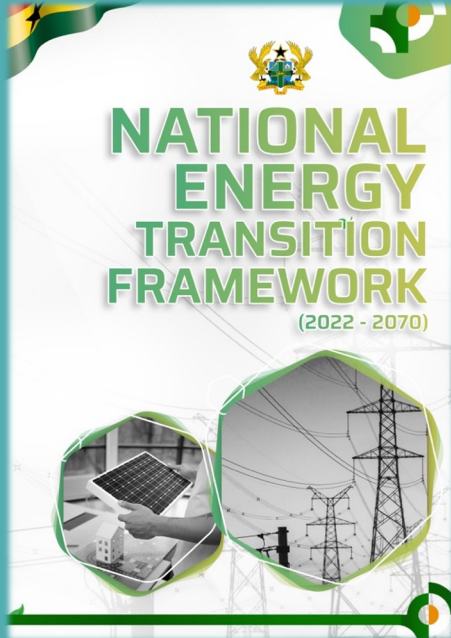
Developing countries require about 1.7 trillion each year to invest, in sustainable energy

Only 544 billion USD was brought in 2022

Increase in annual clean energy investment in selected countries and regions, 2019-2023e



Ghana's Energy Transition Plans



The framework seeks to:

Identify viable pathways for the country to transition towards carbon-neutrality within a secure and efficient energy sector

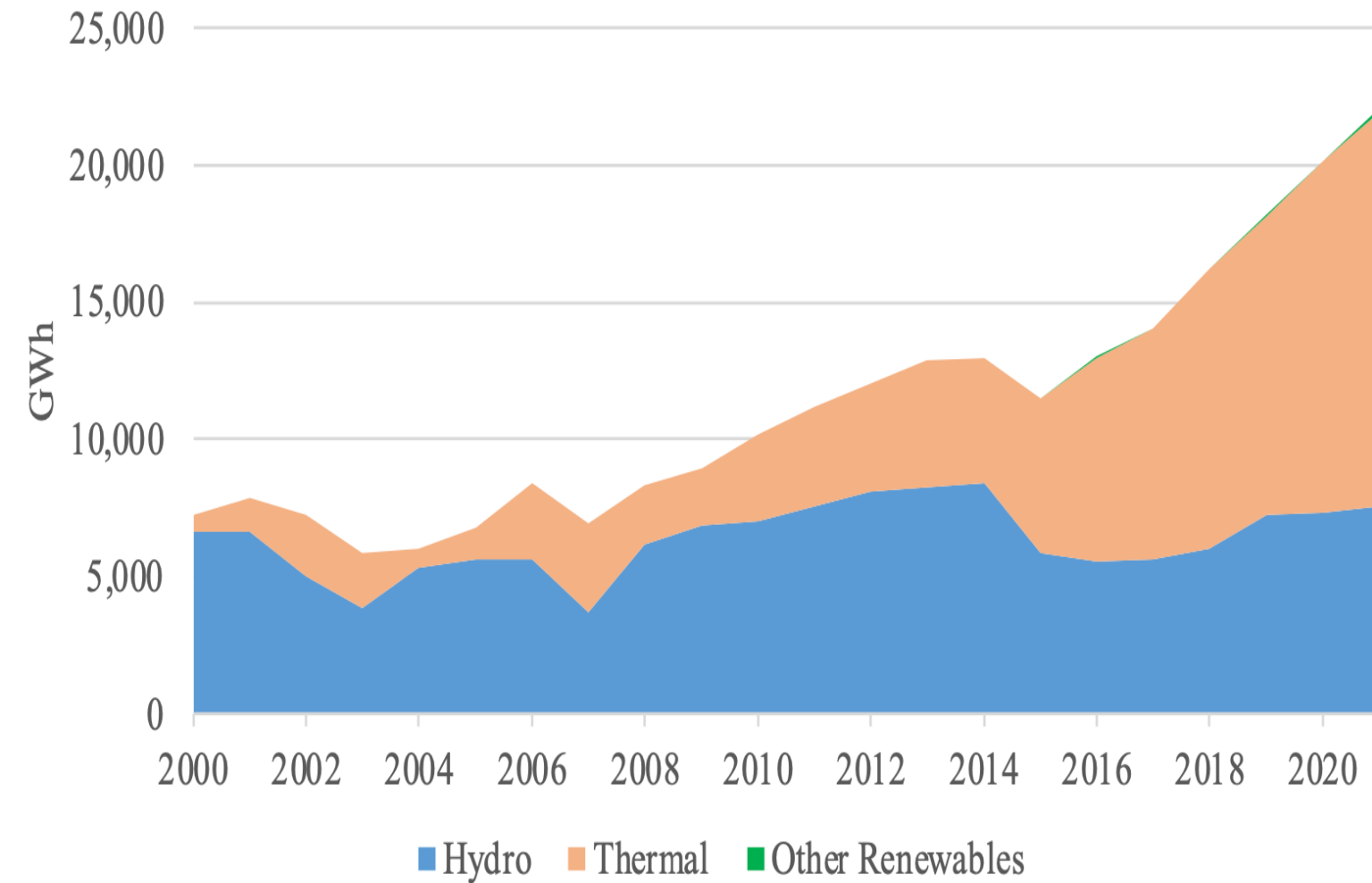
Harness the opportunity for a fair and equitable energy transition as the country relies on carbon-intensive industries for economic growth

Evaluate the impacts of energy transition on the economy (infrastructure, government revenue, jobs, and social development);

Develop medium to long-term targets and policies for achieving a carbon-neutral economy;

Estimate the cost of implementing the framework and identify financing options for the realization of the stated objectives.

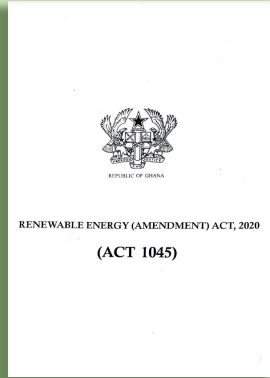
Addresses the financial needs for achieving a net-zero emissions economy by 2060



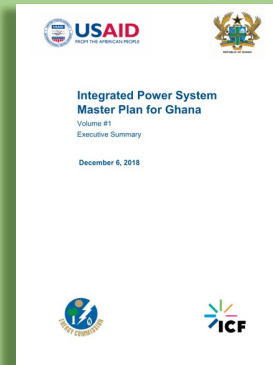
Ghana's Energy Mix

Electricity Generation (GWh)

Policy Frameworks and Regulations

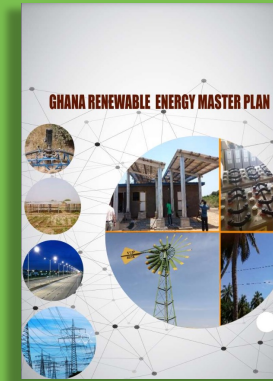


- Legislative framework for advancement of renewable energy
- Establishes the Renewable Energy Authority to supervise its execution
- Offers incentives for renewable energy projects, including tariff guarantees & tax deductions.
- Mandates inclusion of renewable energy technology in the country's energy mix

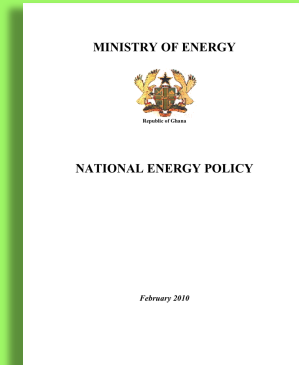


The REMP aims to achieve the following by 2030:

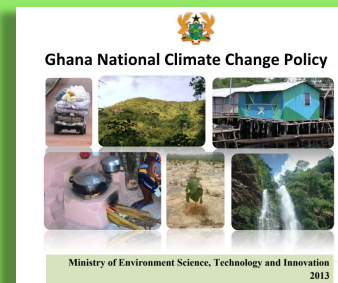
- Increase proportion of renewables in energy mix from 42.5 MW in 2015 to 1363.63 MW
- Reduce dependence on biomass for thermal energy applications
- Provide renewable based decentralised electrification options in 1,000 off-grid communities;
- Promote local content & participation in renewable industry



Aims to identify a long-term Least-Regrets power sector resource plan that will meet Ghana's future electricity demand, through the optimisation of existing and future power plants and other energy systems, as well as transmission capability



Puts emphasis on diversifying energy sources, promoting renewable energy, energy efficiency and conservation, infrastructure development, establishing regulatory frameworks, and stakeholder engagement and capacity building



Seeks to revolutionize four sectors that can address adaptation including energy and infrastructure, natural resources management, agriculture and food security, and disaster preparedness and response

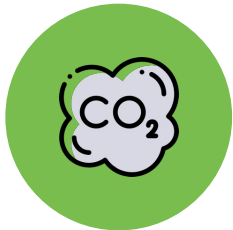
Gaps in Regulatory Framework



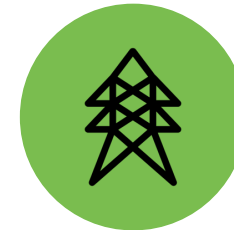
The regulatory framework in Ghana for renewable energy lacks specificity regarding financial incentives and subsidies, such as tax breaks and feed-in tariffs, necessary to attract private sector investment.



The framework lacks adequate provisions for monitoring, reporting, and enforcement mechanisms to ensure compliance with renewable energy regulations and standards.



Ghana's carbon pricing initiatives are still in their infancy. While promising, the carbon levy and green premium capture framework remain limited in scale and scope compared to comprehensive systems seen in other countries

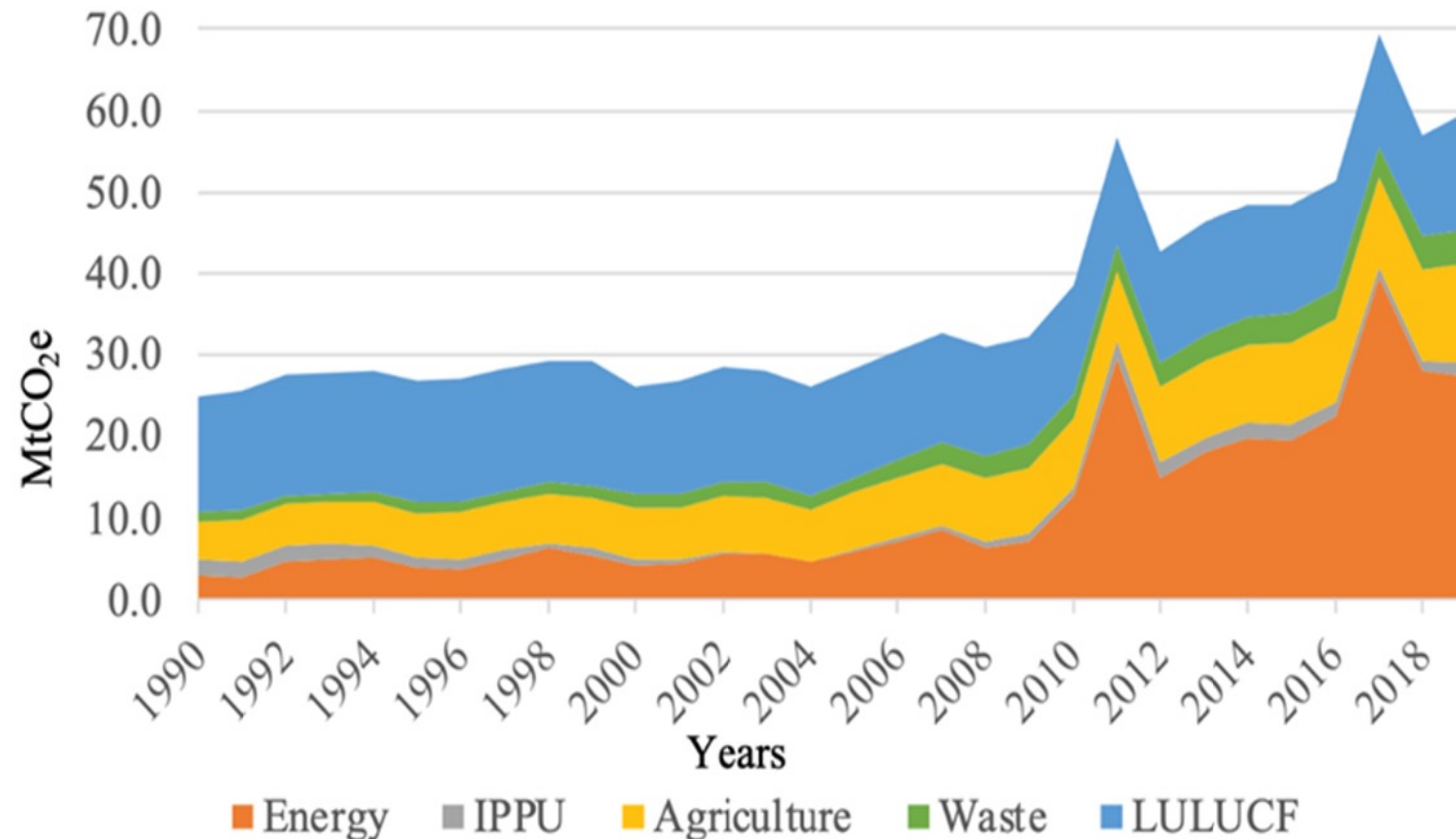


The integration of renewable energy into the national grid requires more detailed provisions, including grid codes, interconnection standards, and transmission infrastructure development.

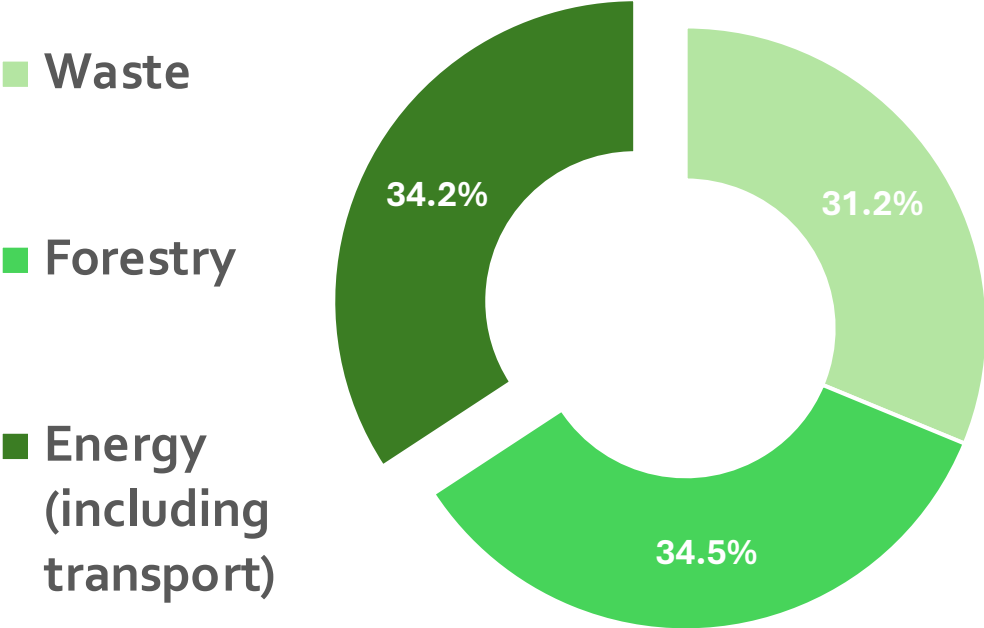
Emissions Reduction Plans and Targets

In 2019, the total national GHG emissions, including Land Use, Land-Use Change and Forestry (LULUCF), reached **59.8 MtCO₂e**

The energy sector has emerged as the leading source of GHG emissions in the country, accounting for **45.7% of total national emissions in 2019**



Emissions Reduction Plans and Targets



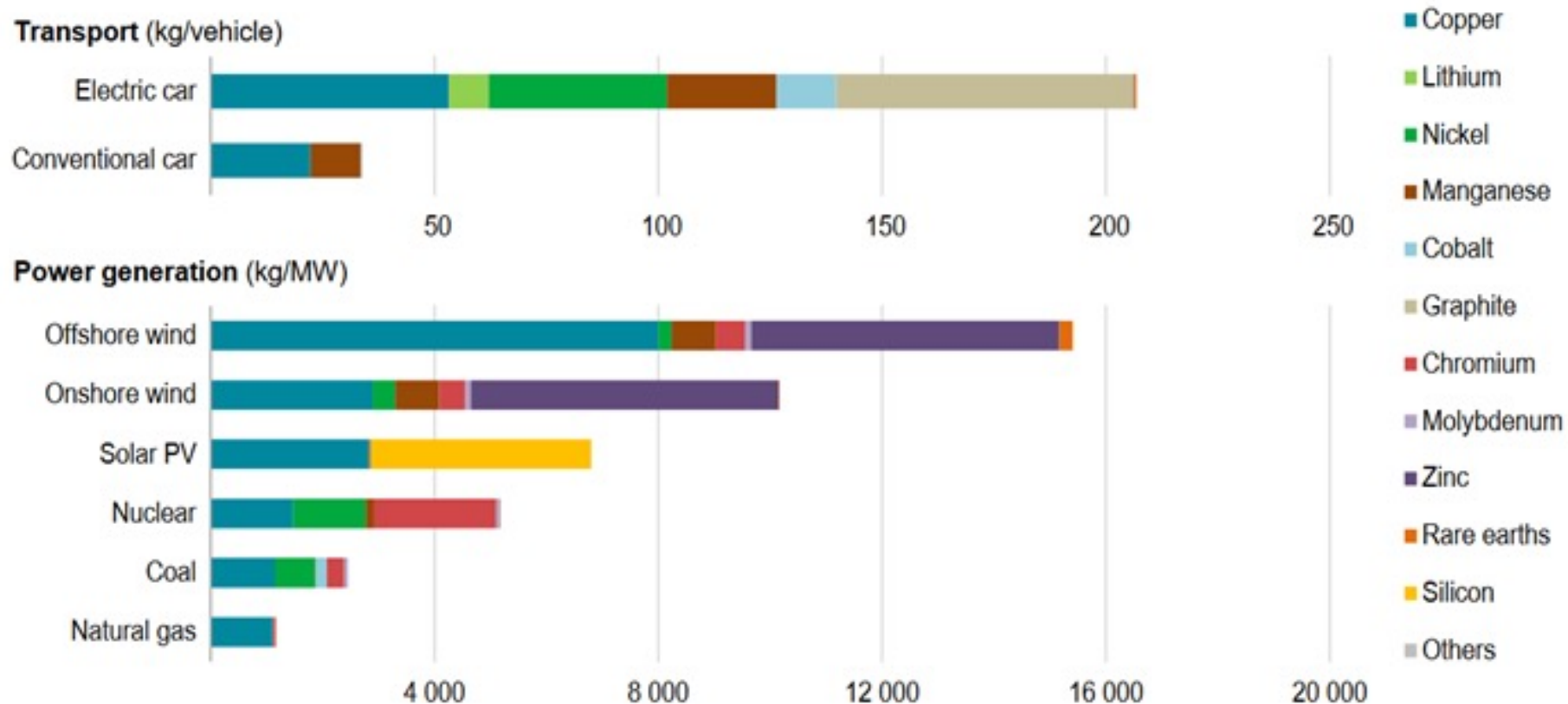
The energy sector including transport is projected to contribute 34.2% of the total national emission reduction target by 2030.

Contributors of Emissions Reductions in the Energy Sector by 2030

NDC Policy Actions (PA)	Climate Objective	GHG saving (kt/yr)
Expansion of inter-and-intra-city transportation modes	Mitigation	109.9
Promotion of energy efficiency in homes, industry, and commerce	Mitigation	1,899.3
Refrigeration and Air Conditioning (RAC)	Mitigation	3,874.2
Sustainable production in Industry	Mitigation	1,480.7
Low carbon electricity generation	Mitigation	4,439.4
Expand the adoption of market-based cleaner cooking solutions	Mitigation	8,438.3
Sustainable Charcoal Production	Mitigation	1,543.0
Promote clean rural households lighting	Mitigation	175.1
Scale up renewable energy penetration by 10% by 2030	Mitigation	1,338.4
Decarbonization of oil and gas production	Mitigation	74.6

Role of Critical Minerals in Ghana's Energy Transition Plan

Minerals Used in Selected Clean Energy Technologies



Financial and Investment Landscape

- **USD 550 billion** in cumulative capital investment (USD 140 billion more than under BAU) is required
 - Transport : **70%**
 - Power & hydrogen: **20%**
 - Industries and buildings: **10%**

TOTAL ANNUAL CAPITAL INVESTMENT REQUIRED BY SECTOR¹, NZE

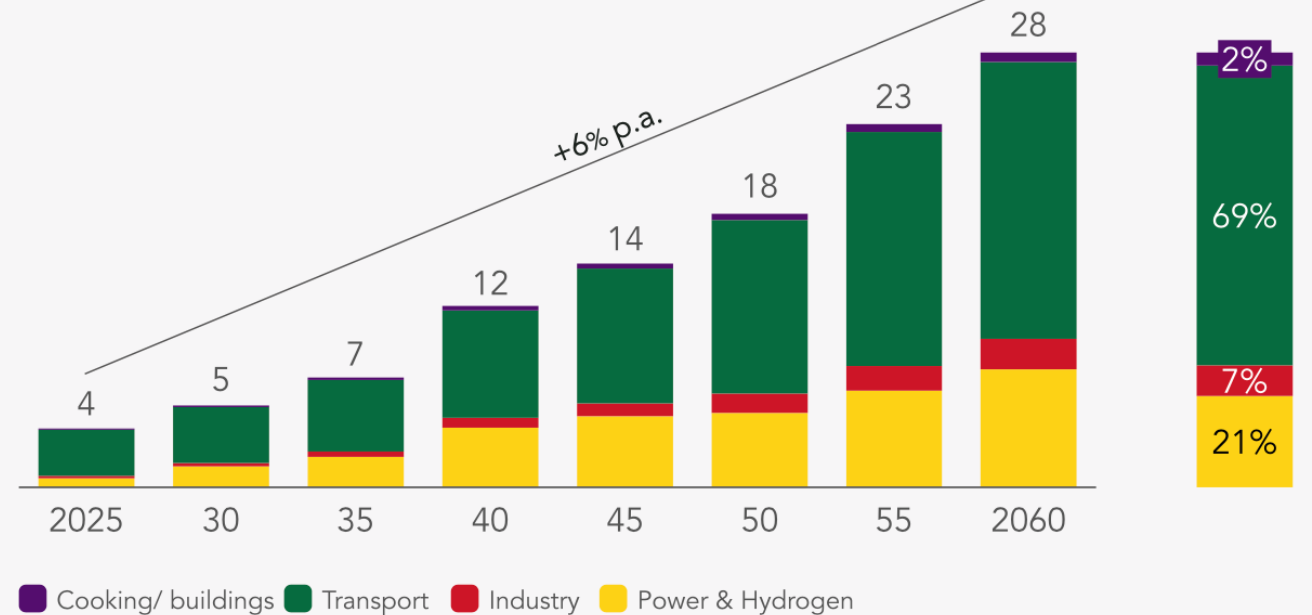
USD bn

Cumulative total, USD bn

138

553

Sector shares
(2025-60)



Funding Sources for the ETP

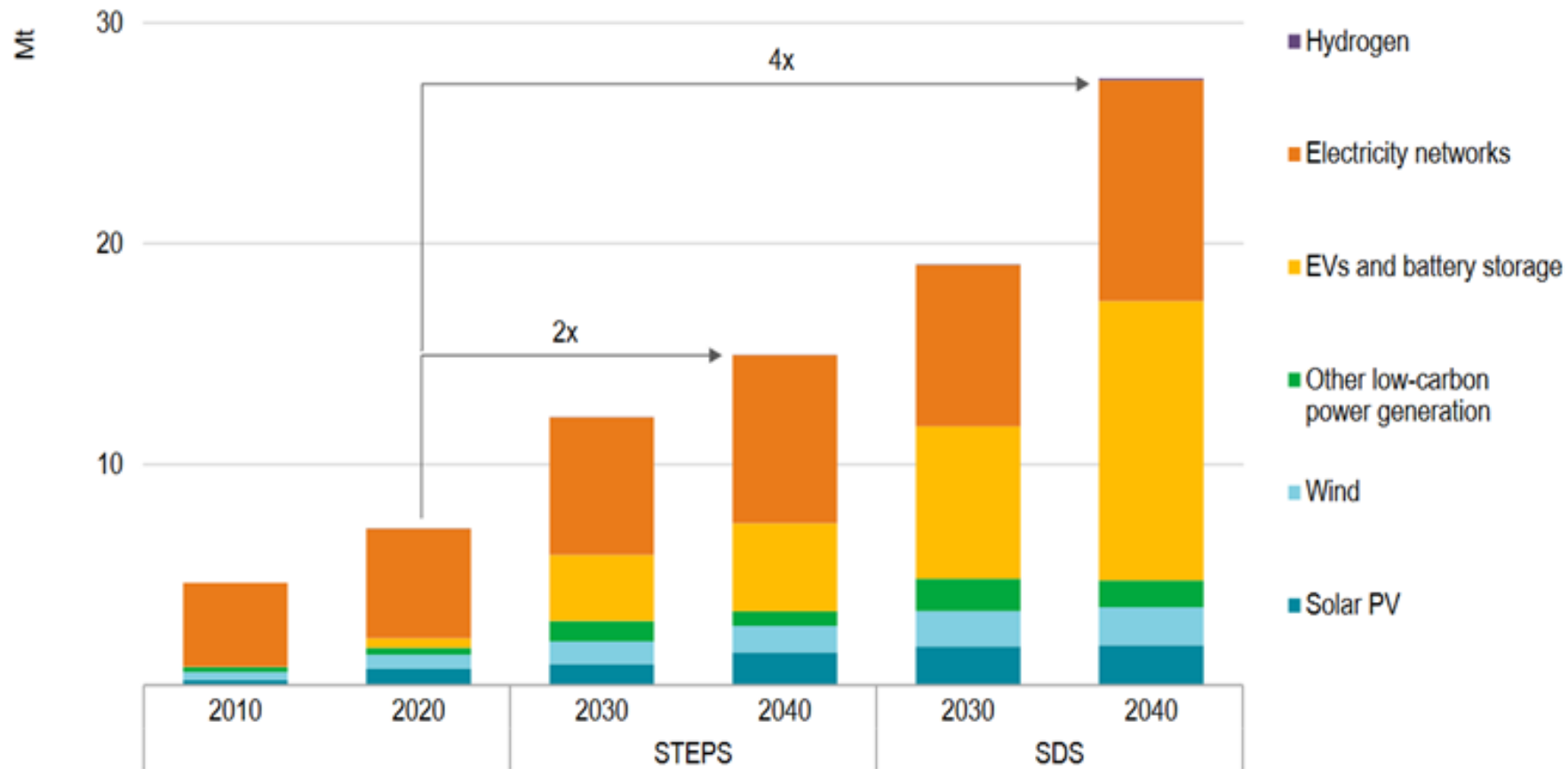
	CORE FINANCE PROVIDERS				PROVIDERS OF DE-RISKING INSTRUMENTS e.g., guarantees/insurance, first-loss-capital, etc.				
	Private sector			Domestic public sector	International institutions				
Actors	Commercial financial institutions	Corporations	Households and individuals	Public institutions	Multilateral DFIs ¹	Bilateral DFIs ¹	National DFIs ¹	Green finance funds	Private foundations
Examples	Ghana Commercial Bank FirstRand Bank BNP Paribas Blackrock Pimco Major pension funds	BHP Billiton Royal Dutch Shell	N/A	Ghana Ministry of Finance	World Bank African Development Bank	French Agency for Development UK FCDO USAID GIZ	Development Bank of Ghana	Green Climate Fund Global Environment Facility Adaptation Fund Clean Technology Fund	Rockefeller Foundation ClimateWorks Foundations Bloomberg IKEA Foundation Bezos Earth Fund

Ghana's Energy Transition Plan is primarily funded through a combination of core finance providers' i.e. (private sector capital and domestic public sector) and de-risking instruments provided.



Role of Critical Minerals in Ghana's Energy Transition Plan

Total Mineral Demand for Clean Energy Technologies by Scenario



Role of Critical Minerals in Ghana's Energy Transition Plan

Ghana's Critical Minerals



Lithium

15 year mining lease granted, estimated life mine of revenues exceeding \$4.8 billion.



Graphite

Castle Minerals announced resource estimate for its Kambale Graphite of 14.4mt @ 7.2% for 1.03mt contained graphite



Silica Sand

ETF does not place significant emphasis on utilizing silica deposits in transition



Iron Ore

ETF advocates for the establishment of refineries to process & add value to before export



Bauxite

ETF advocates for the establishment of refineries to process & add value to before export



Manganese

ETF advocates for the establishment of refineries to process & add value to before export

“

Ghana's Energy Transition Framework advocates for the **establishment of refineries to process and add value to some critical minerals before export** – in line with the mandate of the Ghana Integrated Aluminum Development Corporation (GIADEC) and Ghana Integrated Iron and Steel Development Corp (GIISDEC) to **develop an integrated bauxite, manganese and steel ecosystem** to efficiently harness Ghana's resources

”

Role of Critical Minerals in Ghana's Energy Transition Plan

Exploration & Production Challenges

- The absence of comprehensive geological data on Ghana's critical minerals is a significant technical challenge that requires urgent redress.
- Enforcement of existing legal and regulatory provisions in the small-scale sub-sector has remained a challenge, particularly with the influx of Chinese and the introduction of sophisticated mining equipment, including the Chang fa.

Socio-Economic Impacts

- Ghana can equally leverage its critical mineral deposits to accelerate industrialization and propel socio-economic growth and development.
 - Ghana can utilize Ewoyaa Lithium project and the Automobile Development Policy to become a regional EV production hub, capitalizing on a \$7 trillion market opportunity by 2030
- Exploitation of critical minerals may disproportionately impact women, while incidences of forced child labour may rise

“ While the growth of critical minerals will play a key role in enabling a clean energy transition, if poorly managed, the production and processing of these minerals can also have negative repercussions on the environment and present governance, health and safety and human rights risks. ”

Institutional Readiness for the Transition Power Sector

Main Stakeholders	Role in the Transition	Expansion Strategies in the Framework
MINISTRY OF ENERGY	Sets overall policy direction and manages the power sector.	• Increase the share of renewable energy in the power generation mix. • Modernize grid infrastructure. • Encourage private sector participation.
ENERGY COMMISSION	Regulates and supervises the utilization of energy resources, including renewable energy.	• Licensing, regulation, and supervision of energy resource utilization.
VOLTA RIVER AUTHORITY	Responsible for electricity generation.	• Increase electricity generation capacity from renewable sources. • Promote renewable energy projects.
GHANA GRID COMPANY	Responsible for electricity transmission.	• Modernize grid infrastructure to support renewable energy integration. • Improve transmission capacity and efficiency.
ELECTRICITY COMPANY OF GHANA (ECG)	Main distribution utility, responsible for electricity distribution.	• Improve distribution infrastructure and efficiency. • Enhance electricity access in underserved areas. • Promote clean cooking solutions.
NORTHERN ELECTRICITY DISTRIBUTION COMPANY (NEDCO)	Distributes electricity in Northern Ghana.	• Expand electricity distribution network to cover more areas. • Promote renewable energy adoption in distribution networks.

Institutional Readiness for the Transition Transport Sector

Main Stakeholders	Role in the Transition	Expansion Strategies in the Framework
MINISTRY OF TRANSPORT	Sets policy direction and manages the transport sector.	• Promote sustainable transportation. • Develop electric vehicle infrastructure. • Encourage cleaner fuels.
DRIVER AND VEHICLE LICENSING AUTHORITY (DVLA)	Registers and licenses vehicles.	• Implement regulations and standards for vehicle emissions and efficiency. • Promote electric vehicle adoption.
NATIONAL ROAD SAFETY AUTHORITY (NRSA)	Promotes road safety and reduces accidents.	• Enhance road safety measures and regulations. • Promote eco-friendly transportation practices.

Institutional Readiness for the Transition Oil and Gas Sector

Main Stakeholders	Role in the Transition	Expansion Strategies in the Framework
MINISTRY OF ENERGY	Sets policy direction and manages the oil and gas sector.	• Promote the use of cleaner fuels. • Implement measures to reduce emissions. • Encourage private sector participation.
PETROLEUM COMMISSION	Regulates and manages the utilization of petroleum resources.	• Regulation and oversight of oil and gas activities. • Promote cleaner fuel usage and emission reduction measures.
GHANA NATIONAL PETROLEUM CORPORATION (GNPC)	Responsible for oil and gas exploration and production.	• Invest in cleaner fuel technologies. • Implement emission reduction measures in exploration and production activities.
TEMA OIL REFINERY (TOR)	Refines crude oil.	• Explore cleaner refining technologies. • Promote the use of low-sulfur fuels.
BULK OIL STORAGE AND TRANSPORTATION COMPANY (BOST)	Stores and distributes petroleum products.	• Invest in infrastructure for cleaner fuel storage and distribution. • Implement measures to reduce environmental impact.

Institutional Readiness for the Transition

Environmental Oversight

Main Stakeholders	Role in the Transition	Expansion Strategies in the Framework
MINISTRY OF ENVIRONMENT, SCIENCE, TECHNOLOGY, AND INNOVATION (MESTI)	Sets policy direction and manages environmental issues.	• Strengthen regulatory capacity. • Promote environmental impact assessments (EIAs). • Encourage best practices for environmental management.
ENVIRONMENTAL PROTECTION AGENCY (EPA)	Regulates and manages environmental impacts.	• Monitor and enforce compliance with environmental regulations. • Conduct environmental impact assessments for energy projects.

Institutional Readiness for the Transition - Building & Infrastructure Development

Main Stakeholders	Role in the Transition	Expansion Strategies in the Framework
Ministry of Works and Housing	Sets policy direction and manages building and infrastructure development.	• Promote energy-efficient building design and construction. • Develop sustainable urban infrastructure. • Encourage renewable energy use in buildings.

Socio-Economic Impact of the ETP



Job Creation & Job Erasure

- ETF seeks to encourage more individuals & businesses to invest in small-scale renewable energy projects, thereby fostering the growth of the local industry



Poverty Alleviation & Energy Access

- ETF aims to :
 - Achieve universal access to electricity by 2025,
 - Promote the adoption of clean cooking solutions
 - Reduce reliance on traditional biomass
- Measures to address energy poverty is necessary to ensure that the transition does not exacerbate existing social & economic inequalities

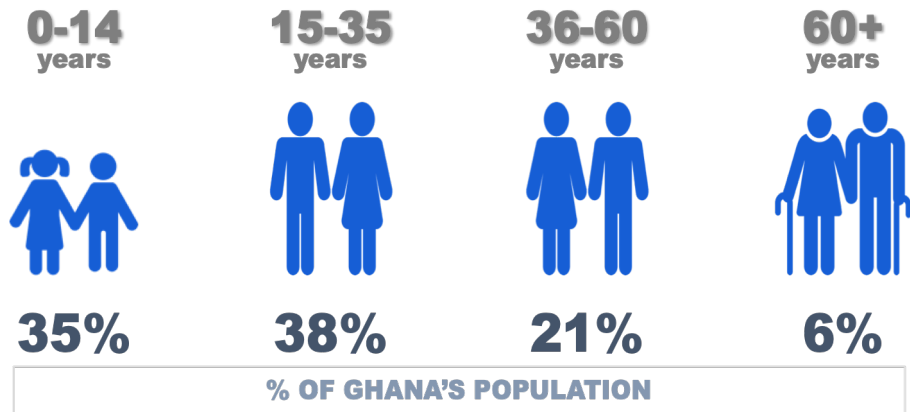


Community & Inclusion

- Engagement of communities at the frontline of the transition is necessary
- Transition frameworks need to incorporate social protection measures to support communities and individuals adversely affected by the transition
- Elaborate strategies for promoting empowering women in the sector, and addressing gender-specific energy needs are missing

Relevance of Youth Engagement in the Transition

Age Demographic Distribution of Ghana



- **Catalyzing Innovation:** To bring fresh perspectives, innovative ideas, and technological savvy in renewable energy technologies, energy-efficient practices, and sustainable development initiatives
- **Building Sustainable Communities:** To foster a sense of ownership and empowerment among young people, enabling them to become active agents of change within their communities.
- **Securing Inter-Generational Equity:** To include them in decision making and ensure that the interests and concerns of future generations are adequately represented..

“ The youth's emergence as leaders during the critical phase of the transition can be transformed from a mere happenstance into a calculated strategy that harnesses emerging technologies and engenders an infusion of fresh perspectives into the governance structures and policy frameworks that will underpin this transition. ”

Investment Opportunities for the Youth

Renewable Energy Entrepreneurship

Production, installation & maintenance of renewable energy systems, catering to both domestic & commercial markets.

Support for youth-led renewable energy startups through targeted funding schemes, incubation programs, and technical assistance.



Energy Efficiency

Providing energy auditing, retrofitting, & consultancy services to help businesses & households reduce their energy consumption & costs

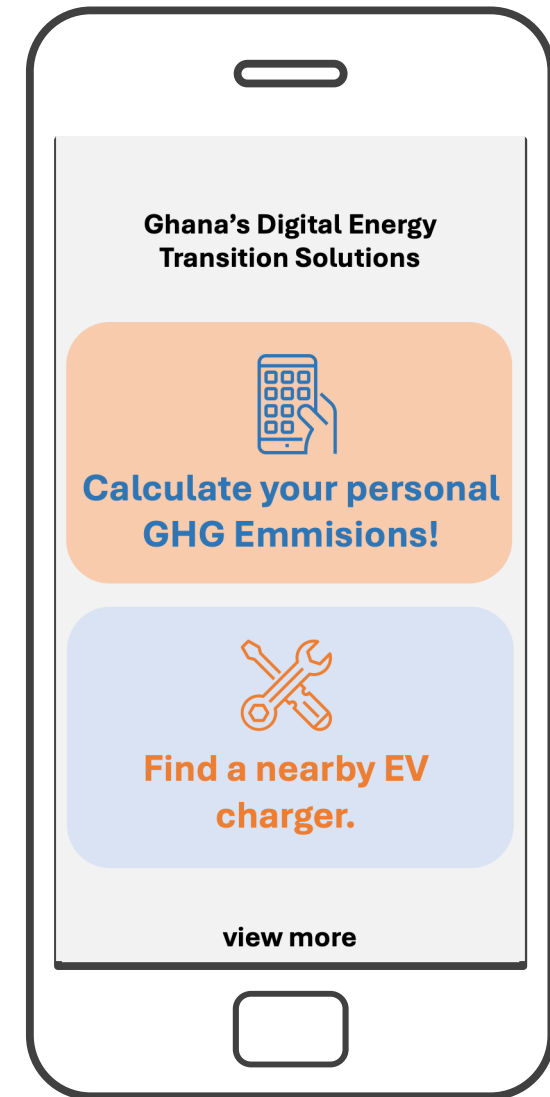
Capacity-building programs & certification schemes can be established to equip youth with necessary skills & credibility to offer these services.

Investment Opportunities for the Youth

Digital Solutions for Energy Management

- ET opens up opportunities for youth with skills in software development, data analytics, & artificial intelligence
- Innovative solutions for energy management, such as smart metering, demand response platforms, & energy trading applications

Government or industry led hackathons, incubation programs, and partnerships with technology companies can nurture youth talent in this area.



Summary of Key Findings

- Ghana's Energy Transition Framework Sets Ambitious Targets, but Implementation Challenges Exist
- Significant Investment Required, with the Power and Transport Sectors Dominating
- Renewable Energy and Energy Efficiency are Key Drivers of Decarbonization
- Transport Sector Electrification and Clean Fuels Offer Significant Emission Reduction Potential
- Critical Minerals Present Economic Opportunities, but Sustainable Exploitation is Crucial
- Socioeconomic Impacts and Stakeholder Engagement are Essential for a Just Transition
- Institutional Capacity Building and Access to International Climate Finance are Key Challenges



Prioritize Realistic and Achievable Targets

Focus on low-hanging fruits and cost-effective measures

- ✓ Accelerate deployment of small-scale, distributed renewable energy solutions, particularly in off-grid and underserved areas.
- ✓ This can include solar home systems, mini-grids, and solar water pumps for agriculture.
- ✓ These solutions can be more cost-effective than grid extensions in remote areas and can provide immediate benefits to communities.

Adjust the timeline & phasing of capital-intensive projects

- ✓ Prioritize the rehabilitation and upgrade of existing power infrastructure, such as transmission & distribution networks, before embarking on extensive expansion projects e.g. solar home systems, mini-grids, & solar water pumps for agric.
- ✓ These solutions can be more cost-effective than grid extensions in remote areas and can provide immediate benefits to communities.

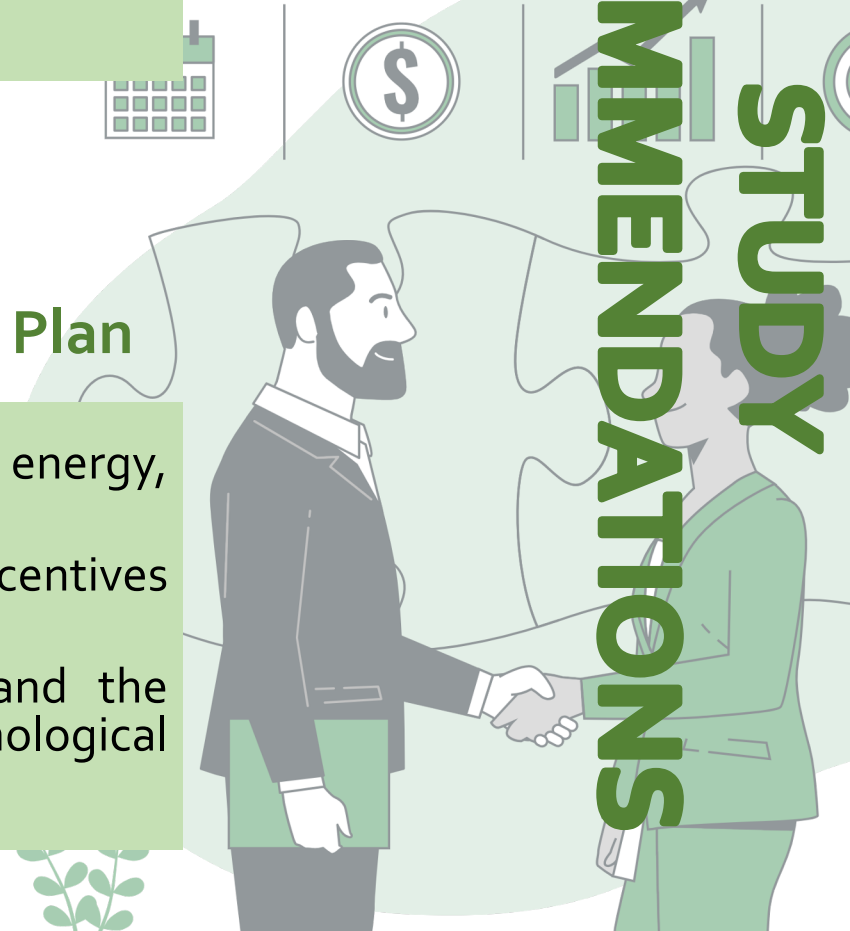
Leverage Partnerships & Innovative Financing Mechanisms

Increase engagement with international development partners, multilateral financial institutions, and climate finance funds to secure concessional financing and grants for priority energy transition projects. This can help bridge the funding gap and reduce the burden on public resources.

Develop a Comprehensive Technology Advancement Plan

- ✓ This plan should outline strategies for promoting R&D in renewable energy, energy storage, electric vehicles & clean cooking solutions
- ✓ Foster collaborations between academia & industry & provide incentives for private sector participation in R&D activities
- ✓ Include measures for technology transfer, capacity building, and the localization of clean energy technologies to enhance technological capabilities and create new economic opportunities

STUDY
RECOMMENDATIONS



STUDY RECOMMENDATIONS

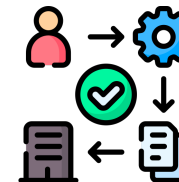
Ensure Efficient Exploitation of Critical Minerals

- ✓ Comprehensive geological assessments to determine the extent & feasibility of critical mineral deposits,
- ✓ Establish a transparent & competitive licensing process
- ✓ Implement robust environmental and social safeguards
- ✓ Prioritize value addition & the development of local processing capabilities to maximize economic benefits



Engage Stakeholders & Address Social Impacts

- ✓ Conduct impact assessments to identify potential social & economic risks, developing targeted support programs for workers in carbon-intensive sectors, & promoting alternative livelihood opportunities.
- ✓ Prioritize investments in engagement, education, skills development, & social protection measures to mitigate the potential adverse impacts of the transition on vulnerable populations



STUDY RECOMMENDATIONS



Enhance Institutional Capacity for Accessing International Climate Funds

- ✓ Establish dedicated units or task forces within key institutions to coordinate and streamline the process of accessing international climate fund
- ✓ Provide training & resources to enhance the technical expertise of staff in proposal development, project design, and financial management.



Establish a Dedicated Energy Transition Fund

- ✓ This fund should be designed to mobilize and allocate financial resources specifically for the implementation of projects and programs outlined in the ETF.
- ✓ The Fund should have clear governance structures, transparent allocation criteria, and robust monitoring and evaluation mechanisms to ensure the effective utilization of resources.

Thank You.

Questions?